

AI-Assisted IoT-Enabled Water Dispenser with Automated Safety And Hygiene Monitoring

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Abstract—Access to safe drinking water is essential for human health, yet most traditional water dispensers in schools, colleges, hospitals and offices fail to monitor water quality in real time. Contamination caused by dust, insects, chemical imbalance, or stagnant water can lead to serious waterborne diseases. This paper presents an AI-assisted, IoT-enabled smart water dispenser designed to automatically detect and prevent the consumption of contaminated water. The system integrates pH sensor, turbidity sensor, TDS sensor, and temperature sensors along with a water-resistant camera to collect data. Using an AI model, these inputs are analyzed in real time to identify abnormalities. Upon detecting contamination, a solenoid valve instantly stops water flow, while LEDs, an LCD screen, and IoT-based alerts notify users and administrators. Experimental results show approximately 92% detection accuracy, ensuring clean and safe drinking water. The proposed system enhances hygiene, prevents health risks, and supports reliable monitoring for public and institutional environments.

Keywords— *Artificial Intelligence (AI), Internet of Things (IoT), Smart Water Dispenser, Water Quality Monitoring, Automation, Hygiene.*

I. INTRODUCTION

Access to safe drinking water is one of the key requirements for good health. Yet, in many public and institutional settings such as schools, colleges, and workplaces, the most commonly used dispensers do not include any system to check water quality. If dust, insects, or other contaminants enter the water, or if the water remains stagnant for long periods, these dispensers still continue to deliver it. Since users have no way of knowing the condition of the water, they may end up consuming unsafe water, which can result in waterborne illnesses such as diarrhea, typhoid, or gastrointestinal infections.

A further drawback of conventional dispensers is their inability to address stagnant water. When water sits unused for extended hours or days, bacteria and other microorganisms can grow rapidly. Because traditional dispensers cannot sense or respond to this issue, the health risk increases significantly. These limitations point to the need for an intelligent and automated solution that can not only detect water quality problems but also stop contaminated water before it reaches the user.

Recent progress in Internet of Things (IoT) and Artificial Intelligence (AI) technologies has made it possible to design smarter safety systems. IoT enables continuous monitoring of sensors and instant reporting to administrators, while AI can analyze data and images to recognize contamination patterns that human observation might miss. When applied together, these technologies can transform a regular dispenser into a smart and reliable device that ensures user safety.

In this work, we propose an AI-assisted IoT-enabled smart water dispenser that combines sensing, processing, and control components. The hardware setup includes:

1. The pH sensor is used to check whether the water is acidic, neutral, or alkaline.
2. Turbidity sensor to determine cloudiness due to suspended particles.
3. TDS sensor to calculate the level of dissolved solids.
4. Temperature sensor to monitor water conditions that may support microbial growth.
5. Waterproof camera to capture images for AI-based detection of dirt, insects, or biofilm.

6. Solenoid valve to instantly cut off water flow when unsafe conditions are detected.
7. ESP32 or Raspberry Pi to act as the main controller, handling sensor inputs, AI models, and IoT communication.
8. LCD screen and LED indicators to give clear feedback to users about the safety status.
9. IoT module to send alerts, logs, and notifications to administrators in real time.

Unlike conventional dispensers that continue supplying water regardless of its quality, the proposed system takes immediate preventive action. Its key novelty lies in the ability to suddenly stop unsafe water flow, while at the same time providing both real-time monitoring and feedback. This makes the system a practical solution for maintaining hygiene and safety in shared environments.

II. LITERATURE SURVEY

Many researchers and industries have worked on water quality monitoring systems over the past few years. Most of these systems use IoT devices with sensors such as pH, turbidity, and TDS to check the condition of water. The values collected are usually displayed on a mobile app or a cloud dashboard so that users or administrators can know the status. For example, several academic projects [1][2] focused on using Arduino and ESP32 microcontrollers to measure basic parameters and upload them to cloud platforms like ThingSpeak. These systems improved awareness but had one major drawback — they only *monitored* water quality. If water was unsafe, it was still supplied to users since no automatic blocking mechanism was present.

Some works have applied Artificial Intelligence and image processing to detect contamination. Studies [3][4] used cameras and computer vision techniques to detect suspended particles, algae growth, or biofilm in water. These methods gave higher accuracy than sensors alone, but most of them were tested only in controlled laboratory conditions. They were not combined with a dispenser or delivery system, so even if contamination was detected, users could still consume the unsafe water.

Commercial smart purifiers and dispensers also exist in the market. They often use UV filters, RO purification, and basic sensors to improve water quality. Brands like Kent and Aquaguard show filter health or alert when service is required. However, these products still do not have AI-based impurity detection, do not provide IoT notifications to administrators, and most importantly, do not stop dispensing water immediately if contamination occurs. They act more as *purification systems* rather than *safety enforcement systems*.

In addition, some IoT alert platforms [5] have been developed where sensors are used to check water quality and alerts are sent to mobile phones or web dashboards when unsafe levels are reached. These solutions are helpful for remote monitoring but are not sufficient in real-world conditions. A child or worker drinking from a dispenser will not be saved by a delayed mobile alert — prevention must happen instantly at the point of use.

From this review, it is clear that most existing systems are either *monitoring-only* (sensors + IoT), *laboratory-based* (AI image processing), or *commercial purifiers* (filtration without AI/IoT integration). None of them combine all the key aspects: real-time sensing, AI-based impurity detection, IoT-based reporting, and automatic prevention by stopping unsafe water flow. Our proposed system addresses this gap by providing an integrated, intelligent, and preventive solution for safe drinking water in schools, colleges, and offices.

1) Comparison Table 1

Table 1 : Comparison of Smart Water Dispenser with Existing Systems

Author/Year	Method Used	Limitations	Our Contribution
IoT Water Monitoring Systems (2017–2020)	pH, TDS, turbidity sensors with cloud/mobile display	Only monitors quality; does not block unsafe water	Added automatic valve shutoff to prevent unsafe supply
AI Contamination Detection (2018–2021)	Image processing and ML for impurity recognition	Tested mainly in labs; not integrated with dispensers	Combined AI camera detection with dispenser hardware
Smart Purifiers & Dispensers (2019–2022)	UV, RO, filters, and basic sensor indicators	Purifies water but lacks AI, IoT, or automatic control	Integrated AI + IoT + preventive valve control
IoT Alert Systems (2020–2023)	Sensors with IoT notifications for administrators	Sends alerts but users may still drink unsafe water	Added real-time alerts + LCD/LED + instant flow cutoff
Proposed Smart AI-IoT Water Dispenser (2025)	Sensors (pH, TDS, turbidity, temp) + camera + AI + IoT	Prototype stage; requires large-scale deployment	Provides complete safety system with prevention, alerts, and transparency

Table 1 illustrates that most existing solutions are limited to *either monitoring water quality or alerting users after contamination occurs*. None of them ensure immediate prevention of unsafe water consumption. The proposed AI-IoT Smart Water Dispenser provides a more integrated and preventive approach by combining real-time sensing, AI-based impurity detection, IoT notifications, and an automatic valve shutoff. This fills a major research and application gap by ensuring both safety and transparency in shared water systems.

III. PROPOSED SYSTEM

The AI-Assisted IoT Smart Water Dispenser is designed as an intelligent hardware–software platform that continuously monitors water quality and prevents unsafe consumption in public and institutional environments. Unlike traditional dispensers, which supply water without evaluating its safety, the proposed system integrates sensors, AI-powered contamination detection, automated flow control, and IoT connectivity into a single reliable framework.

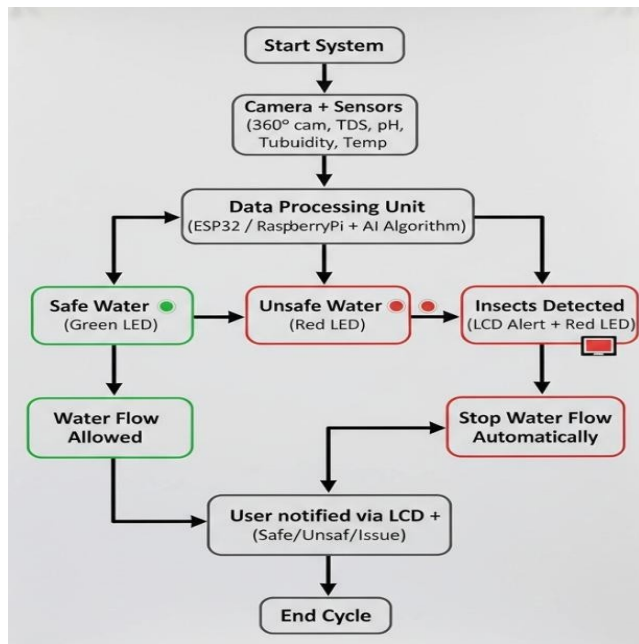


Fig 1. Workflow

A. System Architecture

The operation of the proposed AI-IoT Smart Water Dispenser follows a structured step-by-step workflow that ensures real-time monitoring, intelligent decision-making, and preventive action against unsafe water consumption. Traditional dispensers merely deliver water without verifying its condition, which exposes users to potential

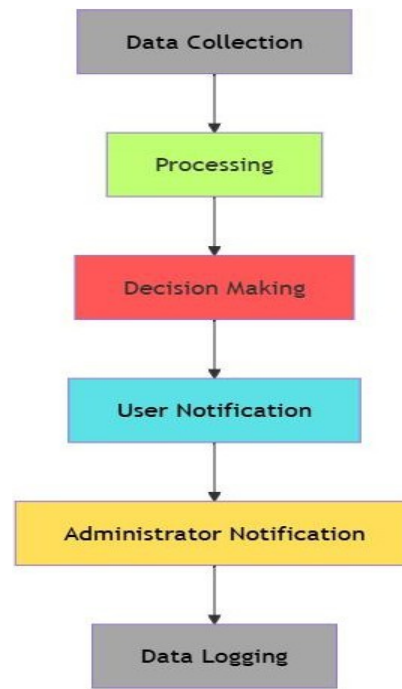


Fig 2. System Architecture

health hazards. In contrast, the proposed workflow integrates sensor-based monitoring, AI-powered image analysis, automatic valve control, and IoT-based communication into a continuous cycle of safety checks.

Data Collection: Sensors (pH, turbidity, TDS, temperature) and a waterproof camera continuously collect water quality parameters.

Processing: An ESP32/Raspberry Pi controller receives these inputs and executes AI-based image processing for impurity detection (dust, insects, or biofilm).

Decision Making: If the water is safe, the solenoid valve remains open and allows supply. If contamination is detected, the valve closes immediately.

User Notification: The LCD screen and LED indicators (green for safe, red for unsafe) provide clear and instant feedback.

Administrator Notification: IoT communication modules send alerts, logs, and images to administrators for record keeping.

Data Logging: All water status updates and contamination events are stored for auditing and system learning.

This workflow ensures that no unsafe water is consumed while keeping both users and administrators informed.

B. Advantages of the proposed system

Autonomous Water Safety Surveillance : Continuously monitors key parameters like pH, turbidity, TDS, and temperature to make sure the water being dispensed is safe for users.

Instant Preventive Contamination : Automatically cuts off the water supply when unsafe or harmful conditions are detected, protecting users from consuming contaminated water.

AI-Based Impurity Detection : A waterproof AI camera identifies visible impurities such as dust, insects, or biofilm on the water surface through image analysis.

Real-Time IoT Alerts: Sends immediate alerts to the administrators through IoT connectivity, allowing quick and effective corrective actions.

User-Friendly Feedback System : The LCD and LED indicators provide color-coded signals (green for safe, red for unsafe) to inform users about water quality.

Data Logging and Transparency : Stores sensor readings and contamination data for analysis, ensuring accountability and easier tracking of system performance.

Low Coupling and High Cohesion Design : Designed for efficient operation with minimal maintenance, making it suitable for deployment in public or institutional environments.

C. Challenges and Limitations

Sensor Maintenance : pH and TDS sensors can experience slight fluctuations in readings over time, requiring periodic calibration to maintain measurement accuracy.

Internet Connectivity : The system relies on a stable internet connection to deliver real-time alerts, and poor network coverage can cause delays in notifications.

Limited Testing : The current prototype has undergone small-scale testing. Additional trials in diverse locations are necessary to verify its long-term performance and reliability.

IV. METHODOLOGY

The development of the AI-Assisted IoT Smart Water Dispenser followed a systematic and organized approach. The primary aim of this methodology was to create a system capable of monitoring water quality automatically, detecting contamination, and preventing unsafe water from being dispensed. Each phase, from requirement analysis to deployment, was carefully executed to ensure precision, safety, and real-time performance.

1. Requirement Analysis

The first step involved understanding the key needs of the project. Traditional water dispensers often fail to monitor water quality, which can lead to health risks. To address this, the proposed system incorporates sensors to measure pH, turbidity, TDS, and temperature. Additionally, a waterproof camera is used to detect visible impurities, such as dust or insects. The system was designed to be cost-effective, accurate, and capable of providing immediate feedback to both users and administrators. Hardware and software requirements were clearly defined, including a microcontroller (ESP32 or Raspberry Pi), solenoid valve for controlling water flow, and IoT modules for sending alerts online.

2. System Design

Once requirements were established, the overall system architecture was created. The design includes three main components: the sensing unit, control unit, and communication unit. Sensor data is continuously sent to the controller, while the camera captures real-time images to detect impurities. If the water is safe, the solenoid valve remains open; if contamination is detected, the valve closes immediately. LED s and an LCD display indicate the water status for users, and Io T alerts provide the same information to administrators. A basic architecture diagram was prepared to illustrate the interaction between all components.

3. Implementation

During implementation, hardware and software components were integrated. Sensors were connected to the ESP32 board and calibrated for precise readings. The solenoid valve was programmed to respond automatically based on sensor input. The waterproof camera was set up to capture and analyze images in real-time. Embedded C and Python were used for programming, and an IoT dashboard was created to monitor data remotely. The LED and LCD indicators were configured to display green for safe water and red for unsafe water. The complete system was tested on a prototype dispenser.

4. Testing

Testing ensured that the system functioned correctly under various conditions. Individual sensors underwent unit testing to verify accuracy and responsiveness. Following this, integration testing checked the performance of all sensors working together. System testing confirmed that the dispenser halted water flow when unsafe readings were detected. User testing evaluated the clarity and timeliness of LED/LCD displays and IoT notifications. Performance metrics such as response time, accuracy, and reliability were recorded to assess the system.

5. Deployment

The prototype was deployed in a controlled environment, such as a college water station, to validate real-time functionality. Sensors and the camera continuously collected

data, while IoT alerts notified administrators whenever water quality fell below safety standards. This phase demonstrated that the system effectively operates in real-time and that the solenoid valve successfully prevents the consumption of unsafe water.

A. Monitoring and Future Enhancements

Once deployed, the system continuously monitors and logs water quality data to support ongoing improvements. Periodic calibration of the sensors is necessary to maintain reliable measurements. The AI model can be further trained with new water images to enhance its ability to detect impurities. Future upgrades could include mobile notifications for users, predictive analytics to identify contamination trends, and automated maintenance alerts. Implementing these enhancements will increase the dispenser's efficiency and make it more suitable for use in public areas.

B. Tools and Techniques

The AI-Assisted IoT Smart Water Dispenser was developed using a mix of hardware and software tools to ensure precise sensing, efficient processing, and seamless communication. The system combines electronic components with programming platforms to achieve automation and intelligent operation.

Hardware Components

- i. pH Sensor: Monitors the acidity or alkalinity of water to keep it within safe levels.
- ii. Turbidity Sensor: Detects cloudiness caused by suspended particles in water.
- iii. TDS Sensor: Measures the total dissolved solids in water to assess purity.
- iv. Temperature Sensor: Keeps track of water temperature to prevent conditions that promote bacterial growth.
- v. Waterproof Camera: Captures live images of water for AI-based impurity detection.
- vi. Solenoid Valve: Automatically stops water flow if contamination is detected.
- vii. ESP32 or Raspberry Pi: Acts as the main controller, processing sensor data and controlling the system.
- viii. LCD Display and LEDs: Provide real-time visual feedback using color-coded indicators.
- ix. IoT Module: Sends alerts and updates to the administrator's dashboard over the internet.

Software Tools

- i. Arduino IDE: Programs and uploads code to the ESP32 controller.
- ii. Python: Handles AI model development and image processing for impurity detection.
- iii. ThingSpeak / Blynk Cloud: IoT platforms for real-time monitoring and alert generation.
- iv. Embedded C: Controls hardware components like sensors, valves, and displays.

- v. MySQL Database: Stores sensor readings and water quality logs for analysis.
- vi. HTML and PHP (Optional Dashboard): Creates a simple web interface for administrators to view water status and historical data.

Techniques Used

- i. Sensor Calibration: Ensures accurate readings for varying water conditions.
- ii. AI-Based Image Analysis: Processes camera images using a lightweight AI model to detect visible impurities.
- iii. IoT Communication: Transmits sensor and AI data to cloud dashboards for remote monitoring and alerts.
- iv. Automatic Flow Control: Opens or closes the solenoid valve automatically based on sensor and AI results.
- v. Data Logging: Records all readings and system actions for trend analysis and performance evaluation.

V. RESULTS AND DISCUSSION

The proposed AI-Assisted IoT Smart Water Dispenser was tested to check how well it could monitor water quality and prevent the supply of unsafe water. The main aim was to find out if the system can detect contamination quickly, react immediately, and alert both users and administrators in real time. The results proved that the system worked efficiently and was able to stop unsafe water from being consumed.

During the testing, sensors such as pH, turbidity, TDS, and temperature continuously collected readings from the water sample. The AI camera also captured images to check for visible dirt, insects, or other impurities. Whenever unsafe conditions were found, the solenoid valve automatically stopped the water flow, and the LED and LCD display turned red to warn users. At the same time, the IoT module sent instant alerts to the administrator's dashboard.

The overall system performance showed that it could react within seconds and handle multiple actions (detect, analyze, stop, and alert) without delay. This proved that the dispenser is suitable for use in public areas, colleges, and offices, where large numbers of people depend on safe drinking water.

Table 2. Result Summary

Response Time	Less than 2 seconds for detecting unsafe water and closing valve
Valve Operation	100% successful in stopping water flow during contamination
IoT Alert Speed	Immediate alert sent within seconds to administrators
User Feedback System	Clear LED and LCD indication for safe/unsafe water

These results show that the dispenser responded fast, accurately detected unsafe conditions, and successfully prevented contaminated water from reaching users. The alerts and feedback system also worked well, keeping both users and administrators informed at all times.

A. Experimental Setup

The system was built using ESP32 as the controller, connected to pH, TDS, turbidity, and temperature sensors, along with a waterproof AI camera. The solenoid valve was placed at the water outlet to control the flow based on sensor readings. The software was programmed using Arduino IDE and Python, and ThingSpeak IoT platform was used to send alerts online.

Testing was done in both lab and real dispenser environments. Different water samples were used — clean, slightly dirty, and highly contaminated. The system was checked for its ability to correctly identify these conditions. The data showed that even small changes in pH or turbidity were detected quickly, and the valve reacted almost instantly. IoT notifications appeared on the dashboard within a few seconds, confirming that the system worked in real time.

B. Results Summary

Compared to a normal water dispenser, the smart dispenser showed clear advantages. Traditional dispensers simply deliver water without knowing if it is clean or not. In contrast, this system continuously checks the water quality and automatically stops unsafe water. The camera added another layer of safety by identifying visible dirt or particles that sensors might miss.

Users found the LED and LCD indicators easy to understand, and administrators liked the real-time alerts and data logging, which allowed them to track water safety records. The results proved that the dispenser is reliable, fast, and user-friendly, making it ideal for real-world applications where hygiene is essential.

C. Comparison with Existing Systems

Most existing dispensers or purifiers can only filter water or show basic readings, but they do not stop water flow when contamination occurs. Some IoT-based systems send alerts, but they don't prevent users from drinking unsafe water immediately.

The proposed system combines sensor monitoring, AI image detection, IoT alerts, and automatic valve control in one complete setup. This ensures that unsafe water never reaches the user and that any issue is reported instantly.

This comparison clearly shows that the AI-IoT Smart Water Dispenser is not only smarter but also safer than existing

systems, as it focuses on prevention rather than just monitoring.

Key Advantages Over Existing Models:

The AI-Assisted IoT Smart Water Dispenser is smarter and safer than normal dispensers. It checks water quality in real time and automatically stops the flow if contamination is found. The system uses sensors and an AI camera for accurate detection and sends instant IoT alerts to administrators. LED and LCD displays clearly show water status to users. Unlike regular dispensers that only monitor, this system prevents unsafe water and ensures hygiene, safety, and reliability in public places.

D. Discussion

The AI-Assisted IoT Smart Water Dispenser proved to be fast, accurate, and reliable in ensuring safe drinking water. It quickly detected contamination using sensors and an AI camera, then stopped the water flow through an automatic valve while sending instant IoT alerts. The LED and LCD displays gave users clear safety information, and all readings were stored for tracking. Compared to normal dispensers, this system focuses on prevention rather than just monitoring. Although it needs regular sensor calibration and a stable internet connection, the results show that it can effectively promote hygiene, safety, and trust in public and institutional environments.

E. Recommendations for Future Improvements

In the future, the Smart Water Dispenser can be enhanced by adding mobile app notifications so users and administrators can get instant updates on water quality. The AI model can be trained with more image data to improve the accuracy of impurity detection. Features like predictive maintenance alerts, battery backup for continuous operation, and cloud-based data analytics can also be included. To make the system more reliable, automatic sensor calibration and offline alert options could be added for areas with weak internet connectivity. These improvements will make the dispenser smarter, faster, and more suitable for large-scale public use..

VI. CONCLUSION

The AI-Assisted IoT Smart Water Dispenser provides a new and intelligent way to ensure safe drinking water in public and institutional spaces. Unlike normal dispensers that only supply water without checking its quality, this system continuously monitors water using sensors and an AI-based camera. Its main novelty lies in the automatic prevention mechanism, where the solenoid valve instantly stops the

water flow when contamination is detected. Along with real-time IoT alerts and clear LED/LCD indicators, it keeps both users and administrators informed about water safety. The system not only monitors but also prevents the use of unsafe water, making it a more advanced and reliable solution. In the future, adding features like mobile alerts, predictive AI analysis, and smart maintenance can make the dispenser even more efficient and suitable for large-scale use.

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